Finding Supply Function Equilibria with Asymmetric Firms
Author(s): Edward J. Anderson and Xinmin Hu
Source: *Operations Research*, Vol. 56, No. 3 (May - Jun., 2008), pp. 697-711
Published by: INFORMS
Stable URL: https://www.jstor.org/stable/25147221
Accessed: 25-03-2026 00:13 UTC

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

INFORMS is collaborating with JSTOR to digitize, preserve and extend access to *Operations Research*

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

OPERATIONS RESEARCH
Vol. 56, No. 3, May–June 2008, pp. 697–711
ISSN 0030-364X | EISSN 1526-5463 | 08 | 5603 | 0697

informs®

DOI 10.1287/opre.1070.0489
© 2008 INFORMS

Finding Supply Function Equilibria with Asymmetric Firms

Edward J. Anderson, Xinmin Hu
Australian School of Business, University of New South Wales, Sydney, NSW 2052, Australia
{eddiea@agsm.edu.au, xinminhu@optusnet.com.au}

Firms compete in supply functions when they offer a schedule of prices and quantities into a market; for example, this occurs in many wholesale electricity markets. We study the equilibrium behaviour when firms differ, both with regard to their costs and their capacities. We characterize strong equilibrium solutions in which, given the other players' supply functions, optimal profits are achieved for every demand realisation. If the demand can be low enough for it to be met economically with supply from just one firm, then the supply function equilibria are ordered in a natural way. We consider equilibria in which, for the highest levels of demand, all but one of the firms have reached their capacity limit. We show that there can be at most one supply function equilibrium with this property. We also propose a new numerical method to find asymmetric supply function equilibria, using piecewise-linear approximations and a discretization of the demand distribution. We show that this approach has good theoretical convergence behaviour. Finally, we present numerical results from an implementation using GAMS to demonstrate that the approach is effective in practice.

Subject classifications: supply function equilibrium; electricity markets; numerical solution of ordinary differential equations; piecewise-linear approximation.
Area of review: Environment, Energy, and Natural Resources.
History: Received January 2006; revisions received August 2006, April 2007; accepted May 2007. Published online in *Articles in Advance* March 11, 2008.

1. Introduction

In many situations, firms compete by offering a schedule of quantities and prices into a market, rather than using a single strategic variable of quantity or price. In this paper, we analyse supply function equilibria (SFE) which occur in this environment. The first complete analysis of such equilibria was carried out by Klemperer and Meyer (1989), who recognized that SFE can exist when there is demand uncertainty, so that firms specify their supply functions before they know the demand. SFE have been used extensively in the analysis of wholesale electricity markets as well as in other environments (Laussel 1992, Akgun 2004). However, there are two significant problems with this approach. First, in general, there may be a whole family of different equilibria, giving rise to an equilibrium selection problem. Second, it has often proved hard to find an equilibrium solution, either analytically or numerically, even when they are known to exist. We make a contribution to both these problems here. We will show that although, in general, there may be a whole family of alternative equilibrium solutions, these will be ordered, and in many cases, there will be a unique solution to the SFE problem with capacity constraints. Moreover, we demonstrate a new robust numerical technique for solving the SFE problem for an oligopoly, and in so doing, we hope to make this approach more useful in practice.

In our model, each firm specifies their supply quantity as a function of the price. A supply function is always required to be nondecreasing, so that the higher the market price, the more commodity a firm is willing to sell. After the simultaneous announcement of the supply function offers, a stochastic demand occurs and the market clears at a single spot price which is then the price paid to each firm for the quantity they supply. In actual markets, there are often other restrictions: for example, on the form of offer allowed.

Wholesale electricity markets have been the main focus for research on SFE, with Green and Newbery (1992) being the first to apply this approach in the England and Wales market. Despite the problems of dealing with equilibria in supply functions, most authors regard this as the most appropriate framework for an equilibrium analysis. Cournot models have been used by a number of authors (e.g., Andersson and Bergman 1995, Borenstein and Bushnell 1999). Cournot models are flexible, can often be solved relatively easily, and there are circumstances where we can expect firms to coordinate on Cournot outcomes. But the case for preferring a supply function approach rests on what happens in actual electricity markets in which firms bid schedules of prices and quantities, rather than quantities alone. For more discussion on some of the modelling issues, see Kahn (1998).

The existence and computation of SFE in general cases has attracted considerable research attention. Klemperer and Meyer (1989) discussed a case with identical convex cost functions for each firm and a concave demand function. They showed the existence of a family of SFE in

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

this case. Electricity demand is almost completely insensitive to spot market price in the short term, and Rudkevich et al. (1998) and Anderson and Philpott (2002a) addressed the case of SFE with stochastic demand which is independent of price, where each firm has an identical cost function. Both papers provide closed-form formulae for strictly increasing SFE when there are no capacity constraints or other operating constraints. Holmberg (2005a) has shown that, when demand is inelastic and capacity constraints bind with a positive probability, there can only be one SFE. Anderson and Xu (2005a) extended the work of Klemperer and Meyer (1989) to deal with hedging contracts, capacity constraints, and price caps, which are all commonplace in electricity markets. However, all of this work is restricted to symmetric solutions in which the firms are identical and each offer the same supply function. Challenges remain for the SFE models when the firms are asymmetrical (having different cost functions) and have operating constraints such as capacity limits.

The simplest form of the asymmetric problem arises when firms have identical cost functions, but have different capacities. Problems of this form, in which marginal cost is constant, are considered by Genc and Reynolds (2004) and Holmberg (2005b). The only case in which an asymmetric SFE can be easily computed is one in which the supply functions are linear (strictly affine). This form of SFE exists when the cost functions are quadratic (linear marginal cost), demand is linear, and capacity constraints are not binding. However, there may also be nonlinear solutions under these conditions, which will be difficult to find analytically. Green (1996, 1999) have shown how affine SFE can be used in analyzing different aspects of the England and Wales market when it operated under a pool system.

It is natural to consider piecewise-linear SFE in which different price ranges have different linear solutions. This approach can be used to approximate more general functional forms. Unfortunately, such solutions cannot always be found because the supply functions determined in this way may decrease in moving from one price interval to the next. Moreover, capacity limits are difficult to incorporate in this framework. Nevertheless, the approach has been used successfully by Baldick et al. (2004) and forms the basis of the numerical approach of Rudkevich (2003).

When we come to consider algorithms for the numerical estimation of SFE, there are a number of options. As we will see later, the equilibrium conditions produce coupled first-order differential equations, and we can use the standard approaches for the solution of such equations. However, there are some difficulties. The primary problem is that a solution can be derived from initial conditions, but infeasibilities in the solution can easily occur, either through a supply function starting to decrease or because the quantity exceeds a capacity limit. Thus, we may need to search among potential initial conditions to locate a feasible solution. Another difficulty arises because the equations are poorly conditioned:ODE solvers may encounter difficulties when the system becomes close to singular, which happens at the lowest point of the supply function where the quantity is close to zero and we expect the price to approach the marginal cost of supply (Baldick and Hogan 2002 give an augmentation of the differential equations to address this problem). Finally, when using the ODE approach, we need to be careful that the solution of the ODE system is actually an equilibrium point rather than being just locally stationary. That is, we need to check that the supply function offered by the ith firm is optimal given the offers from the other firms. Holmberg (2005c) gives a numerical procedure for finding a solution. His approach uses numerical integration to solve the system of ODEs, and searches for feasible solutions by varying the values for the prices at which the capacity constraints are reached for each firm.

There are a number of alternative approaches that have been suggested. For asymmetric firms, Baldick and Hogan (2002) develop an iterative scheme based on piecewise-affine solutions. At each stage, the existing set of supply functions are moved toward the best piecewise-affine response to the other firms' offers. Day and Bunn (2001) use a similar approach to computationally determine equilibrium solutions, but they use a step length of one in their iterative procedure which may be responsible for the cycling that they observe.

Baldick and Hogan (2004) investigate the use of polynomial approximations. They show that there are significant difficulties with this approach, particularly if it is required that the equilibria are stable—by which they mean that a small perturbation by one player will still lead to convergence if each player repeatedly responds according to its best response to the current offers of other players. As shown by Rudkevich (1999), when players are restricted to affine supply functions and use a Cournot adjustment process, there is a rapid convergence to the equilibrium solution.

Rudkevich (2003) shows how a piecewise-linear optimal response can be developed allowing for both horizontal and vertical segments in a grid framework. He also discusses the difficulties of finding an SFE and proposes a method in which each player observes prices and from these develops a piecewise-linear approximation to the supply function of its rivals.

In this paper, we begin in §2 by discussing some fundamentals of the equilibrium model and characterizing the equilibrium solutions that arise. We show that discontinuities in the supply functions can only occur at a price at which one of the other firms begins to supply. We also show that if the lowest demand level has only one firm supplying, then there is an ordering between equilibrium solutions. Moreover, in this case, there can be no more than one equilibrium where all but one of the firms reach their capacity for the highest level of demand. Sec-

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

tion 3 formulates the approximation scheme, in which we take a discretization over demand shocks, rather than over price or quantity. We establish a convergence result for this discretization. Section 4 gives some implementation details for the approximation scheme and presents numerical examples. Section 5 contains concluding remarks.

2. Characterizing Equilibrium Solutions

Our basic model has n firms. Each firm i has a maximal capacity denoted by $\bar{q}_i$. We let $c_i(x)$ be the cost for firm i of producing an amount x, and we assume that $c_i(x)$ is convex and at least twice continuously differentiable. Furthermore, we assume that the marginal cost is always positive, so $c_i'(0) \geqslant 0$ for $i = 1, \ldots, n$. In this section, it is convenient to assume that each firm has a different initial marginal cost of supply, and by reordering the firms if necessary we will suppose that $c_1'(0) < c_2'(0) < \cdots < c_n'(0)$. In the next section when we present our numerical method, we will not need this assumption.

We suppose that demand is given by a function of the form $D(p, \varepsilon) = D(p) + \varepsilon$. The price-sensitive element of the demand, $D(p)$, is smooth and concave, and satisfies $D'(p) < 0$ which implies that the demand function is strictly decreasing. The demand shock ε, which is the stochastic element in the demand, has some distribution with cumulative distribution function $F(\varepsilon)$. Moreover, we assume that this distribution has positive density throughout the interval on which it is defined: specifically, we suppose that there is a density function f with $f(x) > 0$ for $x \in (\varepsilon_{\min}, \varepsilon_{\max})$, where $\varepsilon_{\min}$ is the lowest demand shock that may occur and $\varepsilon_{\max}$ is the largest.

We will assume that there is a given price cap $\bar{p}$, but if there is no explicit price cap, we may take $\bar{p}$ to be the price where $D(\bar{p}) + \varepsilon_{\max} = 0$. At this price, the maximum demand is zero and so the market can only clear at this price in the trivial case that all the supply functions are identically zero. We will suppose that prices are always greater than zero (even though in some electricity markets negative prices can occur).

The supply function for firm i is a function s_i: $[0, \bar{p}] \to [0, \bar{q}_i]$. We will assume that supply functions are nonnegative, nondecreasing, and have a finite number of smooth pieces. Moreover, we assume that $\lim_{p \downarrow p_0} s_i'(p)$ and $\lim_{p \uparrow p_0} s_i'(p)$ exist, where $s_i'(p)$ exists at p. As a result, there is a constant $M > 0$ such that the derivative of all supply functions is bounded by M inside each smooth piece. Supply functions may be discontinuous, and for convenience we assume that they are left-continuous at the points of discontinuity, that is, we define $s_i(p_0) = \lim_{p \uparrow p_0} s_i(p)$, where $s_i(\cdot)$ does not jump at p. However, the results in the rest of this paper would remain essentially unaltered if the supply functions were assumed to be right-continuous, instead of left-continuous.

In fact, once we consider SFE, some forms of "bad" behaviour will be ruled out. For example, Theorem 2 below implies that the supply functions in equilibrium can have jumps only at the marginal cost of zero generation. Note that the assumption of piecewise smoothness is not a restriction in practice, and in fact our assumption is weaker than Klemperer and Meyer (1989) who consider only supply functions that are twice continuously differentiable.

Throughout this paper we, will use the notation $g(x^-)$ and $g(x^+)$ for $\lim_{\delta \downarrow 0} g(x - \delta)$ and $\lim_{\delta \downarrow 0} g(x + \delta)$ when these exist; both quantities are well defined when g is a supply function because this is monotonic. In the same way, we will write $g'(p^-) = \lim_{\delta \downarrow 0} g'(p - \delta)$ and $g'(p^+) = \lim_{\delta \downarrow 0} g'(p + \delta)$ when these exist.

After the supply functions have been chosen, the stochastic demand is realized. Thus, a particular value ε^* of the demand shock ε becomes known and the market clears at a price p^* such that $D(p^*) + \varepsilon^* = \sum_j s_j(p^*)$, and each firm i supplies an amount $s_i(p^*)$. Thus, firm i earns a profit of $p^* s_i(p^*) - c_i(s_i(p^*))$.

We need to be careful, however, if the supply functions are discontinuous (which corresponds to a range of quantities of supply being offered all at the same price). In this case, the market clears at a price p with $\sum_j s_j(p^-) \leqslant D(p) + \varepsilon^* \leqslant \sum_j s_j(p^+)$. If just one supply function, say s_i, is discontinuous at p, then the other firms meet their part of the demand $s_j(p)$, $j \neq i$, and firm i meets the residual demand $D(p) + \varepsilon^* - \sum_{j \neq i} s_j(p)$. When more than one firm has a discontinuous supply function at the same price, then there is no longer a single allocation of demand to firms. Later we will show that in the equilibrium of concern to us this cannot happen, but in case it does happen, we need to specify a rule for sharing demand between firms offering at the same price (see Anderson and Xu 2005b). We will simply assume that there is a preference order between the firms with demand at any price being met first from firm 1, then firm 2, etc (other more realistic sharing rules give the same result).

To find the best response function for firm i, given the supply functions for the other firms s_j, $j \neq i$, we can ask: What choice of price will give the highest profit to firm i? If the market clears at a price p, then the amount supplied by i is given by $s_i(p) = D(p, \varepsilon) - \sum_{j \neq i} s_j(p)$, and this gives a profit of

$$\pi_i(p, \varepsilon) = p\Big[D(p) + \varepsilon - \sum_{j \neq i} s_j(p)\Big] - c_i\Big(D(p) + \varepsilon - \sum_{j \neq i} s_j(p)\Big).$$

If the supply functions $\{s_i(p)\}_{i=1}^n$ in equilibrium are smooth, and there are no other constraints, then the optimal choice of price for firm i is achieved when

$$\begin{aligned}\frac{\partial \pi_i(p, \varepsilon)}{\partial p} &= \Big[p - c_i'\Big(D(p) + \varepsilon - \sum_{j \neq i} s_j(p)\Big)\Big] \\ &\quad \cdot \Big[D'(p) - \sum_{j \neq i} s_j'(p)\Big] + \Big[D(p) + \varepsilon - \sum_{j \neq i} s_j(p)\Big] \\ &= [p - c_i'(s_i(p))]\Big[D'(p) - \sum_{j \neq i} s_j'(p)\Big] + s_i(p) = 0. \quad (1)\end{aligned}$$

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

Thus, we obtain the following first-order (unconstrained) optimality conditions (see also Klemperer and Meyer 1989, p. 1251)

$$s_i(p) = [p - c_i'(s_i(p))]\left[\sum_{j\neq i} s_j'(p) - D'(p)\right], \qquad i = 1, \ldots, n. \quad (2)$$

Note that the constrained version of (2) with constraints $0 \leqslant D(p) + \varepsilon - \sum_{j\neq i}(s_j(p)) \leqslant \bar{q}_i$ is

$$s_i(p) = [p - c_i'(s_i(p)) + \lambda_i - \mu_i]\left[\sum_{j\neq i} s_j'(p) - D'(p)\right], \qquad i = 1, \ldots, n, \quad (3)$$

where the Lagrange multipliers λ_i, μ_i satisfy the conditions

$$\lambda_i \geqslant 0, \quad \mu_i \geqslant 0, \quad \lambda_i s_i(p) = 0, \quad \mu_i[\bar{q}_i - s_i(p)] = 0.$$

If the s_j are continuous at p, but not smooth, then we need to replace (2) with

$$[p - c_i'(s_i(p))]\left[\sum_{j\neq i} s_j'(p^-) - D'(p)\right] \leqslant s_i(p) \leqslant [p - c_i'(s_i(p))]\left[\sum_{j\neq i} s_j'(p^+) - D'(p)\right].$$

If $s_i(p)$ satisfies these inequalities for a given p, then (subject to a global optimality condition) firm i earns the maximum profit possible when the demand shock is the ε which corresponds to this market clearing price. If the condition holds for all p, then the profit for firm i is optimal whatever the demand shock, and consequently whatever the distribution $F(\cdot)$.

The optimal choice for $s_i(p)$ can also be constrained either by the allowable range of prices so the first-order optimality conditions hold only when, at the solution, $0 < p < \bar{p}$. An important case arises when all but one of the supply functions are either zero or at their capacity limits in a range $(p - \delta, p + \delta)$. Then, if s_i is the supply function that is not constrained, its value is given by

$$s_i(p) = -(p - c_i'(s_i(p)))D'(p). \quad (4)$$

In effect, firm i is acting as a monopoly over this range of prices.

There is a further complication which arises from the restriction that the supply functions are nondecreasing. When the best response function has negative slope, then it is not admissible as a supply function, and we are no longer able to find an optimal supply function which satisfies the optimality conditions (2) for each p. In this case, we will need to use a supply function which is constant over an interval. In this case, the choice of supply function is no longer optimal for every ε, and instead we must weigh up losses and benefits for different demand shock realizations. The consequence is that we will seek a supply function that is optimal for the expected profit: the choice of supply function is no longer independent of the demand shock distribution, F, and the first-order optimality conditions given above will not hold. Solving the best-response problems which then arise is more complex, and is fully described in Anderson and Philpott (2002b) and Anderson and Xu (2002).

We say that a supply function $s_i(p)$ is *strongly optimal*, as a response to a set of supply functions $s_j(p)$, $j \neq i$, if for every demand realization, the amount supplied by firm i under this demand realization gives firm i the highest achievable profit subject to constraints on price and capacity. In other words, the situation discussed in the previous paragraph does not occur, so that a strongly optimal supply function remains optimal whatever the distribution of demand shocks, F. We use the term *strong equilibrium* to refer to an equilibrium in which each supply function is piecewise smooth and strongly optimal given the other supply functions. In this paper, we will only consider strong equilibria. It is our belief that other forms of equilibria will exist in very restricted circumstances. This is also supported by the work of Baldick and Hogan (2002) who show that, under relatively mild conditions, a supply function that is constant over an interval cannot be part of an equilibrium.

The set of possible SFE will be constrained according to the range of the supply shocks: in general, there are a smaller set of equilibrium solutions possible when the demand has a wide range of possible values. For a specified supply function equilibrium, we can find the minimum and maximum values of price which occur—we call these $p_{\min}$ and $p_{\max}$, respectively (corresponding to demand shocks $\varepsilon_{\min}$ and $\varepsilon_{\max}$). The values of the supply functions outside the range $(p_{\min}, p_{\max})$ are irrelevant and we cannot say anything about the characteristics of the supply functions outside this range.

We will proceed in stages in our characterization of the SFE. We begin by showing that in a strong equilibrium, no supply is offered at prices below marginal cost, which translates into the following statement about supply functions.

Lemma 1. *In a strong equilibrium,*

$$s_i(p) = 0 \quad \textit{if and only if} \quad p_{\min} < p \leqslant c_i'(0);$$
$$c_i'(s_i(p)) < p \textit{ for } c_i'(0) < p \leqslant p_{\max}.$$

Proof. Consider the first statement. Suppose that there is a price p_0 with $s_i(p_0) > 0$ and $p_0 \leqslant c_i'(0)$. Consider letting p approach p_0 from below. From our assumption on left continuity of the supply functions, we may assume that the optimality condition (3) holds as p approaches p_0. Using the bound on the derivatives of $s_j'(p)$, $j \neq i$, and because $D'(p) > D'(p_0)$, we have $s_i(p) \leqslant [p - c_i'(s_i(p)) - \mu_i] \cdot [(n-1)M - D'(p_0)]$. Because $c_i'(s_i(p_0)) \geqslant c_i'(0)$, the term

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

$p - c_i'(s_i(p)) - \mu_i$ approaches a number which is negative or zero. Because $s_i(p)$ approaches $s_i(p_0)$, we have a contradiction, which establishes that $s_i(p) = 0$ if $p \leqslant c_i'(0)$.

On the other hand, suppose that there is some price p_0 with $s_i(p_0) = 0$ and $p_0 = c_i'(0) + \delta$ for some $\delta > 0$. We let p approach p_0 from below. Observe that $c_i'(s_i(p))$ approaches $c_i'(s_i(p_0)) = c_i'(0)$ and so $p - c_i'(s_i(p)) > \delta/2$ for p close enough to p_0. Moreover, it is easy to see that there will be some lower bound; call it $\kappa > 0$, on $\sum_{j\neq i} s_j'(p) - D'(p)$ for p close enough to p_0 because $D'(p_0) < 0$. Thus, we can use the optimality condition (3) to show that $s_i(p) \geqslant \kappa\delta/2$ as $p \to p_0$ which contradicts $s_i(p_0) = 0$. Hence, we have established the if and only if statement of the first part of the lemma.

To establish the second part of the lemma, we use a similar argument. Suppose that $p_0 > c_i'(0)$, so that $s_i(p_0) > 0$. Considering p approach p_0 from below and using the optimality condition (3), we have $p_0 > c_i'(s_i(p_0))$. □

The next step is to show that the solutions are continuous (with some specific exceptions). The argument is roughly as follows. Suppose that $s_i(p_A)$ jumps from q_a to q_b. The consequence is that there is range of demands at which the market will clear at p_A. Hence, there is a nonzero probability of a demand occurring in this range and there is a jump in the profit for some other firm by offering at a price just below p_A in comparison to just above p_A. This is enough to show that this is not an equilibrium (this type of situation is described in more detail in Anderson and Xu 2005b). The result below makes this more precise.

THEOREM 2. *In a strong equilibrium, if $s_i(p_A^-) < s_i(p_A^+)$ with $p_A < \bar{p}$, then* (a) *all firms k, $k \neq i$, for which $s_k(p_A) > 0$ are at their capacity limit, i.e., $s_k(p_A) = \bar{q}_k$, and* (b) *there is some firm $j \neq i$ with $p_A = c_j'(0)$.*

PROOF. We begin by showing that there cannot be a strong equilibrium with two supply functions discontinuous at the same price. Suppose otherwise, and let $\{s_j(p)\}_{j=1}^n$ be a strong equilibrium in which both s_i and s_k are discontinuous at p_A. Now consider a demand shock ε with $\sum s_j(p_A^-) - D(p_A) < \varepsilon < \sum s_j(p_A^+) - D(p_A)$ so that there is an ambiguity in the amount supplied by i and k. Then, from Lemma 1, both $c_k'(s_k(p_A)) < p_A$ and $c_i'(s_i(p_A)) < p_A$, and so both firms would prefer to have the largest possible supply. The sharing rule gives preference to one of the firms, say i. Then, firm k will improve its profits by choosing a slightly lower price than p_A, which contradicts the fact that this is a strong equilibrium.

Thus, to prove part (a), we suppose that we have $s_i(p_A^-) < s_i(p_A^+)$ and $0 < s_k(p_A) < \bar{q}_k$, with s_k continuous at p_A. Thus, from Lemma 1, we have $c_k'(s_k(p_A)) < p_A$. Because supply function s_i has a jump at p_A, the price will be p_A for a range of values of demand shock ε, specifically for ε between $\sum s_j(p_A^-) - D(p_A)$ and $\sum s_j(p_A^+) - D(p_A)$. For ε in this range, we let

$$\Delta(\varepsilon) = D(p_A) + \varepsilon - \sum_j s_j(p_A^-) > 0.$$

We select a demand shock ε_A in this range with $\Delta(\varepsilon_A)$ small enough so that $\Delta(\varepsilon_A) < \bar{q}_k - s_k(p_A)$ and $c_k'(s_k(p_A) + \Delta(\varepsilon_A)) < p_A$ (using the continuity of c_k').

We claim that p_A is not an optimal price for firm k for demand shock ε_A given the other firms' supply functions. That is, $\{s_j(p)\}_{j=1}^n$ does not form a strong supply function equilibrium. In fact, if firm k chooses a price $p_A - \delta$, instead of p_A, with this demand shock, then its supply changes from $s_k(p_A)$ to

$$D(p_A - \delta) + \varepsilon_A - \sum_{j\neq k} s_j(p_A - \delta).$$

As $\delta \to 0$, this quantity approaches $s_k(p_A) + \Delta(\varepsilon_A)$ (which is less than $\bar{q}_k$ by construction). Hence, the change in profit for firm k in moving from p_A to $p_A - \delta$ is

$$\begin{aligned}
&(p_A - \delta)\Big[D(p_A - \delta) + \varepsilon_A - \sum_{j\neq k} s_j(p_A - \delta)\Big] \\
&\quad - c_k\Big(D(p_A - \delta) + \varepsilon_A - \sum_{j\neq k} s_j(p_A - \delta)\Big) \\
&\quad - p_A s_k(p_A) + c_k(s_k(p_A)) \\
&= p_A\Delta(\varepsilon_A) + [c_k(s_k(p_A)) - c_k(s_k(p_A) + \Delta(\varepsilon_A))] + O(\delta) \\
&\geqslant [p_A - c_k'(s_k(p_A) + \Delta(\varepsilon_A))]\Delta(\varepsilon_A) + O(\delta)
\end{aligned}$$

because c_k is convex. Therefore, this change in profit is positive for δ chosen small enough. This contradicts the fact that this is a strong equilibrium. Therefore, $s_i(p)$ cannot have a jump at p_A.

To prove part (b), we show first that there must be at least one firm $j \neq i$ with $0 < s_j(p)$ as p approaches p_A from above or $s_j(p) < \bar{q}_j$ as p approaches p_A from below (i.e., s_j increases from zero to the right of p_A or s_j hits its capacity bound at p_A). If not, then there is some interval $(p_A - \delta, p_A + \delta)$ over which the other firms offer a constant quantity (either 0 or $\bar{q}_j$). So, over this interval firm i chooses a price p to maximize $\pi_i(p, \varepsilon) = p[D(p) + \varepsilon - \Delta] - c_i(D(p) + \varepsilon - \Delta)$, where $\Delta = \sum_{j\neq i} s_j(p)$. This expression has first and second derivatives with respect to p given by

$$\pi_i' = pD'(p) + D(p) + \varepsilon - \Delta - c_i'(D(p) + \varepsilon - \Delta)D'(p),$$
$$\pi_i'' = [p - c_i'(s_i(p))]D''(p) + 2D'(p) - c_i''(s_i(p))(D'(p))^2.$$

Because $D' < 0$, $D'' \leqslant 0$, $c_i'' \geqslant 0$, it is easy to see that $\pi_i'' < 0$ and so π_i is concave and will have a single optimizing choice of p. Because the functions D and c_i are smooth, this choice of p will depend continuously on the demand shock ε, which shows that the supply function will also be continuous in this interval, giving a contradiction.

So now we suppose that there is no firm $j \neq i$ with $0 < s_j(p) < \bar{q}_j$ as p approaches p_A from above, and therefore there is some set of firms (excluding i), with $s_j(p) < \bar{q}_j$ as

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

p approaches p_A from below and $s_j(p_A) = \bar{q}_j$. We call this set $I(p_A)$.

Using (2), we can write the expressions for the values of s_i either side of the jump as follows:

$$s_i(p_A^-) = -[p_A - c_i'(s_i(p_A^-))]\Big[D'(p_A) - \sum_{j\in I(p_A)} s_j'(p_A^-)\Big], \quad (5)$$

$$s_i(p_A^+) = -[p_A - c_i'(s_i(p_A^+))]D'(p_A). \quad (6)$$

Because c_i is convex, $c_i'(s_i(p_A^-)) \leqslant c_i'(s_i(p_A^+))$, and moreover $s_j'(p_A^-) \geqslant 0$. Using $p_A - c_i'(s_i(p_A^+)) > 0$, we can deduce

$$-[p_A - c_i'(s_i(p_A^-))]\Big[D'(p_A) - \sum_{j\in I(p_A)} s_j'(p_A^-)\Big] \geqslant -[p_A - c_i'(s_i(p_A^+))]\,D'(p_A),$$

but this contradicts our assumption that s_i jumps up at p_A.

Hence, we have shown that there is a firm $j \neq i$ with $0 < s_j(p)$ as p approaches p_A from above, and $s_j(p_A^+) = 0$ by (a). Using Lemma 1 establishes that $p_A = c_j'(0)$ as required. □

One implication of Theorem 2 is that the supply function of each firm in a strong SFE is continuous if all the firms have identical cost functions (see Klemperer and Meyer 1989, and for identical cost functions, but different capacity limits, see Holmberg 2005b).

Given a strong SFE $\{s_i(p)\}_{i=1}^n$, then, by Theorem 2, there are only finitely many price points at which some $s_i(\cdot)$ is not continuous. We define $H(p) = \{i\colon 0 < s_i(p) < \bar{q}_i\}$, the set of the unconstrained firms at p. Because an individual firm can only enter $H(p)$ at one price and then leave it at a higher price, if $H(p_1) = H(p_2)$, then H must be constant over the interval $[p_1, p_2]$. We define $H_-(p_0) = \lim_{p_0 > p \to p_0} H(p)$ and $H_+(p_0) = \lim_{p_0 < p \to p_0} H(p)$. It is easy to see that $H_+(p)$, $H_-(p)$ are well defined. Moreover, $H(p)$ is a subset of a set with finitely many elements, and so there are a finite set of points at which the set function H changes. Note, however, that we could have $H_+(p)$, $H_-(p)$, and $H(p)$ all different if some supply function s_i hits its capacity limit at the same price, $c_j'(0)$, at which s_j leaves zero. Using Lemma 1, we can see that $i \notin H(c_i'(0))$, which is the price at which i enters $H(p)$. Hence, $H(p) \subset H_-(p)$.

At a smooth point p of a strong SFE $\{s_i(\cdot)\}$, we can rewrite (2) as

$$\sum_{j\in H(p)} s_j'(p) - s_i'(p) = \frac{s_i(p)}{p - c_i'(s_i(p))} + D'(p) \quad \text{for } i \in H(p). \quad (7)$$

This is well defined except when $p - c_i'(s_i(p)) = 0$. From Lemma 1, we know that this only occurs when $p = c_i'(0)$.

Summing this over $i \in H(p)$, we have

$$(m-1)\sum_{j\in H(p)} s_j'(p) = \sum_{i\in H(p)} \frac{s_i(p)}{p - c_i'(s_i(p))} + mD'(p),$$

where $m = |H(p)|$, the number of firms in $H(p)$. Thus, when $m > 1$, we have

$$s_i'(p) = \frac{1}{(m-1)}\Big(\sum_{j\in H(p)} \frac{s_j(p)}{p - c_j'(s_j(p))} + D'(p)\Big) - \frac{s_i(p)}{p - c_i'(s_i(p))} \quad \text{for } i \in H(p). \quad (8)$$

It will be convenient to write $h_i(p, x)$ for the expression $x/(p - c_i'(x))$, so we can rewrite (8) as

$$s_i'(p) = \frac{1}{(m-1)}\Big(\sum_{j\in H(p)} h_j(p, s_j(p)) + D'(p)\Big) - h_i(p, s_i(p)) \quad \text{for } i \in H(p). \quad (9)$$

Lemma 3. *In a strong equilibrium, each supply function s_i has continuous derivatives except at prices $p \in Q$, where Q is the set of prices at which a firm either starts to supply or reaches its capacity limit, i.e.,*

$$Q = \{p_X\colon s_j(p_X) = \bar{q}_j \text{ and } s_j(p) < \bar{q}_j \text{ for } p < p_X\} \cup \{c_1'(0), \ldots, c_n'(0)\}.$$

Proof. By assumption, each s_i has continuous derivatives except at a finite set of points. Suppose that one of these points p_z is not in the set Q. Then, from Theorem 2, each supply function is continuous at p_z. Moreover, $H_+(p_z) = H_-(p_z)$. Because the supply functions are smooth at prices p which approach p_z from either the left or right, Equation (9) applies as p approaches p_z. Because the s_j are all continuous and the set H does not change, this implies that $s_i'(p_z^-) = s_i'(p_z^+)$, which is a contradiction. Thus, the only points at which the derivative of an s_i is discontinuous are points in Q. □

Now suppose that p_A and p_B are adjacent points in Q, and we know $s_i(p_A^+)$, $i = 1, 2, \ldots, n$. Will this be enough to determine the solution throughout the interval (p_A, p_B)? To answer this question, we have to know whether the solutions are well behaved near p_A. We shall address this question in a series of lemmas. (The proofs are given in the online companion to this paper, available as part of the online version that can be found at http://or.pubs.informs.org/ecompanion.html). The first lemma shows that where an ordering between two different equilibria exists at some price, that ordering persists (at least while the supply functions involved are continuous).

Lemma 4. *Let $\{s_i(p)\}_{i=1}^n$ and $\{\tilde{s}_i(p)\}_{i=1}^n$ be two strong SFE, with corresponding sets of unconstrained firms $H(p)$ and $\tilde{H}(p)$. Suppose that there is some price p_0 with $s_i(p_0) \geqslant \tilde{s}_i(p_0)$ for all i with strict inequality for $i \in \tilde{H}(p_0) \neq \varnothing$. If both $\{s_i(p)\}_{i=1}^n$ and $\{\tilde{s}_i(p)\}_{i=1}^n$ are continuous in the range (p_0, p_1), then, for $p \in (p_0, p_1)$, $s_i(p) \geqslant \tilde{s}_i(p)$ for all i with strict inequality for $i \in \tilde{H}_-(p)$.*

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

PROOF. See the online companion.

We will say that a set of different equilibria is an ordered family if any two members satisfy the conditions of Lemma 4 from the point at which they differ, i.e., there is some price p_0 with $s_i(p) \geqslant \tilde{s}_i(p)$ for all i with strict inequality for $i \in \tilde{H}(p)$, for each $p_0 < p \leqslant \min\{p_{\max}, \tilde{p}_{\max}\}$.

The next result concerns the uniqueness of the strong supply function equilibrium at prices $p = c_l'(0)$ for $l = 2, \ldots, n$.

LEMMA 5. *If two strong SFE $\{s_i(p)\}_{i=1}^n$ and $\{\tilde{s}_i(p)\}_{i=1}^n$ have the same right-hand limit at $p_0 = c_l'(0)$ for some $2 \leqslant l \leqslant n$, so $s_i(p_0^+) = \tilde{s}_i(p_0^+)$, $i = 1, 2, \ldots, n$, then $\{s_i(p)\}_{i=1}^n$ and $\{\tilde{s}_i(p)\}_{i=1}^n$ are identical on a range $(p_0, p_0 + \Delta)$ for some $\Delta > 0$.*

PROOF. See the online companion.

The next lemma states that if SFE differ and are ordered, they must each have two or more unconstrained firms.

LEMMA 6. *Let $\{s_i(p)\}_{i=1}^n$ and $\{\tilde{s}_i(p)\}_{i=1}^n$ be two different strong SFE, with corresponding sets of unconstrained firms $H(p)$ and $\tilde{H}(p)$. Let $p_0 \neq c_j'(0)$ for all $j = 1, \ldots, n$ be chosen so that $p_0 < \min\{p_{\max}, \tilde{p}_{\max}\}$, and that $\{s_i(p_0)\}_{i=1}^n$ and $\{\tilde{s}_i(p_0)\}_{i=1}^n$ are defined and not equal. Suppose that $s_i(p_0) \geqslant \tilde{s}_i(p_0)$ for all i, with strict inequality for all $i \in \tilde{H}(p_0)$. Then, both $\tilde{H}(p_0)$ and $H(p_0)$ have two or more elements.*

PROOF. See the online companion.

Now we come to the main result describing the equilibria that may exist.

THEOREM 7. *If*

$$-D(c_1'(0)) < \varepsilon_{\min} < -D(c_2'(0)), \tag{10}$$

then any strong SFE is part of an ordered family, and only the lowest (smallest offers at any given price) can have the property that all but one of the firms reach their capacity limits prior to the maximum price.

PROOF. Condition (10) simply means that at price $c_1'(0)$, there is always some demand (i.e., even when $\varepsilon = \varepsilon_{\min}$), but when the price reaches $c_2'(0)$, there may be no demand for the lowest demand shock levels. Therefore, any equilibrium solution has $c_1'(0) < p_{\min} < c_2'(0)$.

Suppose that there are two equilibria $\{s_i(p)\}_{i=1}^n$ and $\{\tilde{s}_i(p)\}_{i=1}^n$. We take p_V to be the supremum of prices at which these equilibria are the same. The two equilibria cannot diverge except at a point where $p = c_k'(0)$ for some k because otherwise, the ODE defined by (9) is locally Lipschitz which implies a unique solution through a given point. Also, from Lemma 5, if none of the supply functions has a jump at p_V, then we still have uniqueness. Hence, $p_V = c_k'(0)$ for some k and at least one of the equilibria has a jump at p_V. Because the two solutions are identical at prices lower than p_V, they both have just one firm in the set $H(p_V)$, say i. Swapping the labels for the two equilibria if necessary, we may suppose that $s_i(p_V^+) > \tilde{s}_i(p_V^+)$, and hence that $s_k'(p_V^+) > \tilde{s}_k'(p_V^+)$. So the conditions for Lemma 4 apply for p_0 just above p_V. We will show that these conditions continue to apply while both SFE are defined (i.e., up to a price $\min(p_{\max}, \tilde{p}_{\max})$).

Let p_W be the supremum of prices above p_V at which the ordering holds. Suppose that $p_W < \min\{p_{\max}, \tilde{p}_{\max}\}$. By the result of Lemma 4, some element of one of the equilibria $\{s_i(p)\}_{i=1}^n$ or $\{\tilde{s}_i(p)\}_{i=1}^n$ has a jump at p_W. Moreover, if only $s_i(\cdot)$ jumps up at p_W for some i, then it is easy to see that the ordering conditions of Lemma 4 continue to apply. Hence, there must be some $\tilde{s}_i(p_W)$ which is discontinuous, and from Theorem 2, $\tilde{H}(p_W) = \{i\}$. But applying Lemma 6 to $p_W - \delta$ with sufficiently small $\delta > 0$, we have at least two elements in $\tilde{H}_-(p_W)$, and so there must be some j with $\tilde{s}_j(p) < \bar{q}_j$ for $p_W > p \to p_W$ and $\tilde{s}_j(p_W) = \bar{q}_j$. But by Lemma 4, we have $s_j(p_W) > \tilde{s}_j(p_W)$, which gives a contradiction.

We need to show that $\{s_i(p)\}_{i=1}^n$ cannot have the property of all but one of the firms reaching their capacity limits prior to the maximum price. Suppose that $p_{\max} > \tilde{p}_{\max}$. Then,

$$\begin{aligned} D(\tilde{p}_{\max}) + \varepsilon_{\max} &= \sum_{i=1}^n \tilde{s}_i(\tilde{p}_{\max}) < \sum_{i=1}^n s_i(\tilde{p}_{\max}) \leqslant \sum_{i=1}^n s_i(p_{\max}) \\ &= D(p_{\max}) + \varepsilon_{\max}, \end{aligned}$$

which contradicts D decreasing, and hence $p_{\max} \leqslant \tilde{p}_{\max}$.

Now suppose that $s_i(p) = \bar{q}_i$, for $i \neq k$ for some k, for $p \in (p_{\max} - \delta, p_{\max})$. Then, for this choice of p,

$$\begin{aligned} 0 = h_k(p, s_k(p)) + D'(p) &> h_k(p, \tilde{s}_i(p)) \\ &+ D'(p_0) \geqslant h_k(p, \tilde{s}_i(p)) - \sum_{j \in \tilde{H}(p_0),\, j \neq i} \tilde{s}_j'(p) + D'(p_0) = 0, \end{aligned}$$

which is a contradiction. This establishes the result. □

In the case that the lowest level of demand involves more than one firm, so $D(c_2'(0)) + \varepsilon_{\min} > 0$, then we can no longer deduce that the equilibria belong to an ordered family.

Our numerical experiments show that typical problems either have a range of equilibria each of which has at least two firms below their capacity limits for prices below the maximum, or they have a single equilibrium with all but one firm reaching their capacity limits prior to the maximum price. However, as is illustrated in Figure 1, it is possible for a single problem to have both types of equilibria. The figure shows an equilibrium (solid lines) in which firm 2 reaches its capacity limit, and another equilibrium (dashed lines) in which neither firm reaches its capacity limit. This latter equilibrium will be one of a family of nearby equilibria.

Note that all of our results are for the case of strictly decreasing demand curves (an assumption that was used in the proof of Lemma 1). When demand is insensitive to price, a strong equilibrium must always have at least two

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

Figure 1. Possible multiple equilibria when all firms but one have capacity limits.

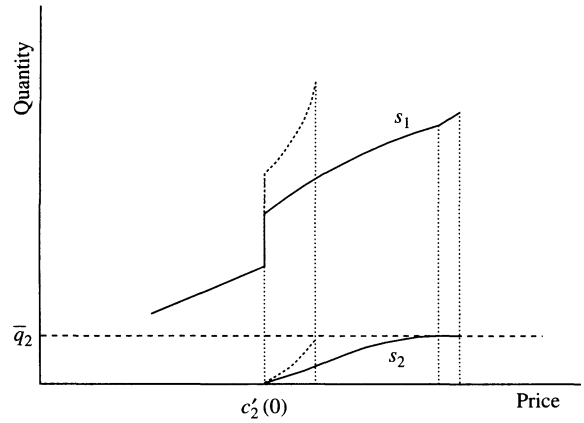
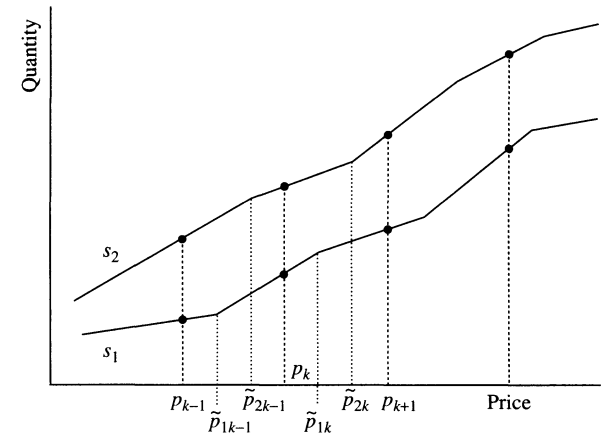


Figure 2. Construction of piecewise-linear supply functions.



elements in the set of unconstrained firms $H(p)$ because in a monopoly situation, prices are set as high as possible in the best response (see (1)). However, it can be shown that the ordered family result still holds with each strong SFE having $s_1(p) = 0$, $p < c_2'(0)$, and $s_1(c_2'(0)) \geqslant D + \varepsilon_{\min}$.

3. Approximation of Supply Function Equilibria

The results we have given allow us to describe the equilibrium solutions that can occur. Now we turn to the numerical calculation of an asymmetric equilibrium. Most researchers have approached this problem through a discretization over prices. We will take a different approach and look instead at an approximation based on a discretization over possible demand shocks.

Thus, we approximate the demand shock distribution using a demand shock profile, $\Omega_\varepsilon = \{\varepsilon_1, \ldots, \varepsilon_K\}$ for some given positive integer K. We may choose to take the values $\varepsilon_1 \leqslant \varepsilon_2 \leqslant \cdots \leqslant \varepsilon_K$ as given by $\varepsilon_k = F^{-1}((k-1)/(K-1))$, $k = 1, \ldots, K$. This will correspond to an assumption that demand is equal to $D(p) + \varepsilon_k$ with probability $1/K$ for each $k = 1, \ldots, K$. However, because we are seeking strong equilibria, the results are independent of the distribution function F, and hence do not depend on the probability with which each of the ε_k occur.

We suppose that the market clears at the price p_k when the demand shock is ε_k, and so

$$\sum_{i=1}^{n} s_i(p_k) = D(p_k) + \varepsilon_k.$$

We write q_{ik} for the amount that firm i is dispatched at this price and β_{ik} for the slope of the supply function for firm i at this point (i.e., $\beta_{ik} = s_i'(p_k)$). We will construct the supply functions in such a way that they are differentiable at p_k, $k = 1, 2, \ldots, K$, so that these slopes are well defined.

We will consider supply functions that are piecewise linear, with separate pieces corresponding to each of the possible demand shocks ε_k. Thus, the kth piece of the supply function for firm i passes through the point (p_{ik}, q_{ik}) at a slope β_{ik}. (It is convenient to write the price here as p_{ik} rather than p_k because we wish to think of the choice of (p_{ik}, q_{ik}) being made by firm i.) To define a supply function, we require that two consecutive segments of a supply curve, applying at p_{ik} and p_{ik+1}, intersect at some point between p_{ik} and p_{ik+1}. This simplifies the representation of equilibrium conditions discussed later (e.g., (14)). This is illustrated in Figure 2. In fact, to ensure that p_{ik} is a smooth point of the piecewise supply curve, we require that the intersection point is strictly inside the interval (p_{ik}, p_{ik+1}). Hence, we require that there exist $\tilde{p}_{ik}$ for $k = 1, \ldots, K-1$ such that

$$q_{ik+1} + \beta_{ik+1}(\tilde{p}_{ik} - p_{ik+1}) = q_{ik} + \beta_{ik}(\tilde{p}_{ik} - p_{ik}), \qquad p_{ik} < \tilde{p}_{ik} < p_{ik+1}. \tag{11}$$

Note that it is possible to have two consecutive lines coincide. In this case, there exists infinitely many points satisfying (11). However, the supply curves constructed below are independent of the choices of such points.

Given (p_{ik}, q_{ik}) and $\beta_{ik} \geqslant 0, k = 1, \ldots, K$, satisfying $p_{ik} < p_{ik+1}, q_{ik} \leqslant q_{ik+1}$ and (11) for $k = 1, \ldots, K-1$, a supply function for firm i is constructed as follows:

$$s_i(p) = \begin{cases} q_{i1} + \beta_{i1}(p - p_{i1}), & 0 \leqslant p \leqslant \tilde{p}_{i1}, \\ q_{ik} + \beta_{ik}(p - p_{ik}), & \\ \quad \tilde{p}_{ik-1} \leqslant p \leqslant \tilde{p}_{ik}, \ k = 2, \ldots, K-1, & \\ q_{iK} + \beta_{iK}(p - p_{iK}), & \tilde{p}_{iK-1} \leqslant p \leqslant \bar{p}. \end{cases} \tag{12}$$

Firm i determines its optimal price, given the other firms' supply functions $s_j(p)$ for $j \neq i$, by solving the following

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

profit-maximization problem for demand shock ε_k:

$$\begin{aligned}\underset{p_{ik}}{\text{maximize}}\ & \left[D(p_{ik})+\varepsilon_k-\sum_{j\neq i}s_j(p_{ik})\right]p_{ik}\\ &\quad -c_i\left(D(p_{ik})+\varepsilon_k-\sum_{j\neq i}s_j(p_{ik})\right)\\ \text{such that}\ & 0\leqslant p_{ik}\leqslant\bar{p},\\ & 0\leqslant D(p_{ik})+\varepsilon_k-\sum_{j\neq i}s_j(p_{ik})\leqslant\bar{q}_i.\end{aligned}\tag{13}$$

The optimal choice of q_{ik} follows from the market-clearing condition at price p_{ik}: i.e., $q_{ik}=D(p_{ik})+\varepsilon_k-\sum_{j\neq i}s_j(p_{ik})$.

Note that the objective function problem for (13) is concave on each of the intervals where $\sum_{j\neq i}s_j(p)$ is smooth. A global maximum can be found by comparing the maximum values of the objective function on each of these finitely many pieces.

We aim to approximate a continuous piecewise-smooth SFE when firms face an uncertain demand which follows a continuous distribution. In fact, as we will show in computational examples, our method will work well for any strong SFE, which as we have already shown, will have at most finitely many jumps. We begin by assuming that the equilibrium price at each of the demand shocks occurs at a smooth point of the supply function (12). So we assemble the equilibrium conditions for $\{(13)\}_{i=1}^n$ for each demand shock realization in the following set of equilibrium conditions (where we now make use of the fact that we are interested in solutions with $p_{ik}=p_k$, $i=1,\ldots,n$):

$$\begin{aligned}&q_{ik}-(p_k-c_i'(q_{ik}))\left(\sum_{j\neq i}\beta_{jk}-D'(p_k)\right)+\lambda_{ik}-\mu_{ik}=0,\\ &\sum_{j=1}^n q_{jk}=D(p_k)+\varepsilon_k,\\ &q_{ik+1}-q_{ik}-\beta_{ik+1}p_{k+1}+\beta_{ik}p_k+(\beta_{ik+1}-\beta_{ik})\tilde{p}_{ik}=0,\\ &\qquad\qquad\text{void when } k=K,\\ &p_k<\tilde{p}_{ik}<p_{k+1},\quad \beta_{ik}\geqslant 0,\\ &0\leqslant p_k\leqslant\bar{p},\quad 0\leqslant q_{ik}\leqslant\bar{q}_i,\\ &\lambda_{ik}\geqslant 0,\quad \mu_{ik}\geqslant 0,\quad \lambda_{ik}(\bar{q}_i-q_{ik})=0,\quad \mu_{ik}q_{ik}=0,\\ &i=1,\ldots,n,\quad k=1,\ldots,K,\end{aligned}\tag{14}$$

where the pieces $q_{ik}+\beta_{ik}(p-p_k)$ and $q_{ik+1}+\beta_{ik+1}(p-p_{k+1})$ intersect at $\tilde{p}_{ik}$, $k=1,\ldots,K-1$ for $i=1,\ldots,n$, and λ_{ik},μ_{ik} are the Lagrange multipliers of the constraints on q_{ik}.

A solution to conditions (14) will determine a piecewise-linear approximation to an SFE. This gives an alternative to a direct numerical solution of the ODE system (2). A basic ODE numerical method will use an approximate forward or backward integration: there is no equivalent to this integration in the approach we propose. Our method avoids making a fixed discretization of the price range as would be the case with a numerical ODE method. Moreover, our approach does not involve any procedure for choosing initial values of the supply functions.

An important advantage of our method, as we show in the theorem below, is that the pairs of prices and quantities of a solution of (14) will lie on the equilibrium supply curves, rather than being an approximation to the curve. To make this more precise, suppose that there exists an equilibrium solution with supply functions $\{s_i^*(p)\}_{i=1}^n$. We would like to show that there is a solution to (14) with the property that $q_{ik}=s_i^*(p_k)$, $i=1,\ldots,n$, $k=1,\ldots,K$ (and then we would expect the slope β_{ik} to approximate the slope of the tangent to s_i^* at p_k).

In some cases, there may be a difficulty due to the nonexistence of a piecewise-linear approximation to s_i^* where the slopes of the pieces match the tangents of s_i^* at certain points (p_k,q_{ik}). For example, if the equilibrium solution was given by the supply function $s_i^*(p)=c+(p-p_0)^3$, and p_1,p_2 (determined by $\varepsilon_1,\varepsilon_2$) were to be two points symmetric about p_0, then the tangent lines of $s_i^*(p)$ at these two points do not intersect. These sorts of problems can only occur when there is a change of sign in the second derivative of s_i^*, and can always be resolved by adding or changing the choice of points ε_k.

THEOREM 8. *Let $\{s_i^*(p)\}_{i=1}^n$ be a strong SFE on $[0,\bar{p}]$. Then, for K large enough, there exists a solution ε_k, p_k, q_{ik}, $\tilde{p}_{ik}$, λ_{ik}, μ_{ik}, $i=1,\ldots,n$, $k=1,\ldots,K$ to (14), such that $D(p_k)+\varepsilon_k=\sum_j s_j^*(p_k)$, $q_{ik}=s_i^*(p_k)$, $i=1,\ldots,n$, $k=1,\ldots,K$. Moreover, there is some number K_0, independent of K, such that $\beta_{ik}=s_i^{*\prime}(p_k)$ for $i=1,\ldots,n$, except for at most K_0 values of k.*

PROOF. We begin by supposing that the supply functions s_i^* are smooth for each $i=1,\ldots,n$. Given a choice of $p_1<p_2<\cdots<p_K$, we define the corresponding demand shock ε_k such that $D(p_k)+\varepsilon_k=\sum_j s_j^*(p_k)$. By assumption, for $k=1,\ldots,K$, p_k is a global optimal solution to (13) for each firm i in the following form:

$$\begin{aligned}\underset{p}{\text{maximize}}\ & \left[D(p)+\varepsilon_k-\sum_{j\neq i}s_j^*(p)\right]p\\ &\quad -c_i\left(D(p)+\varepsilon_k-\sum_{j\neq i}s_j^*(p)\right)\\ \text{such that}\ & 0\leqslant p\leqslant\bar{p},\\ & 0\leqslant D(p,\varepsilon_k)-\sum_{j\neq i}s_j^*(p)\leqslant\bar{q}_i.\end{aligned}$$

The first-order optimality conditions of this problem, when we substitute $q_{ik}=s_i^*(p_k)$ and $\beta_{ik}=s_i^{*\prime}(p_k)$, gives the first two equation sets in (14). Moreover, because s_i^* is monotonic increasing, $\beta_{ik}\geqslant 0$. The assumptions on s_i^* imply $0\leqslant p_k\leqslant\bar{p}$. Because $0\leqslant s_i^*(p)\leqslant\bar{q}_i$, we have $0\leqslant q_{ik}\leqslant\bar{q}_i$, so it only remains to check the conditions involving $\tilde{p}_{ik}$.

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

Consider the continuous function g, defined by

$$g(z) = q_{ik+1} - q_{ik} - \beta_{ik+1}p_{k+1} + \beta_{ik}p_k + (\beta_{ik+1} - \beta_{ik})z. \tag{15}$$

We will be finished if we can show that there is a zero of this function strictly between p_k and p_{k+1}. Now,

$$g(p_k) = [s_i^*(p_{k+1}) + (p_k - p_{k+1})s_i^{*\prime}(p_{k+1})] - s_i^*(p_k),$$
$$g(p_{k+1}) = s_i^*(p_{k+1}) - [s_i^*(p_k) + (p_{k+1} - p_k)s_i^{*\prime}(p_k)].$$

Thus, $g(p_k)$ is the difference, at p_k, between the tangent line to s_i^* taken at p_{k+1} and the curve s_i^* (and similarly for $g(p_k)$). If s_i^* is convex (but not affine) on the interval (p_k, p_{k+1}), then $g(p_k) > 0 > g(p_{k+1})$. If s_i^* is concave (and not affine) on this interval, then $g(p_k) < 0 < g(p_{k+1})$. In either case, there is a point strictly between p_k and p_{k+1} where g is zero (and the same is true if s_i^* is affine in the interval). With an arbitrary choice of ε_k, the only case in which we may not be able to find a point $\tilde{p}_{ik}$ in the required range is when there is a change of sign in the second derivative of s_i^* strictly between p_k and p_{k+1}. In this case, we can move the p_k slightly (equivalent to changing ε_k) or add a new price point p_k (equivalent to adding a new demand shock ε_k) so that p_k is at a point where $s_i^{*\prime\prime}(p_k) = 0$. Hence, we can arrange to have either concavity or convexity for each supply function s_i^*, in all the intervals (p_k, p_{k+1}).

Now consider the case where the equilibrium supply functions (as in Lemma 3) have finitely many (isolated) points at which they are not smooth and possibly not continuous. First, suppose that there is a point, p_z, at which the supply function is continuous but not smooth. We will show that by taking p_k and p_{k+1} approaching p_z from the left and right, respectively, the condition involving $\tilde{p}_{ik}$ will hold. We take $p_k = p_z - \delta$ and $p_{k+1} = p_z + \delta$ and g_δ to be the function defined in (15) with this choice of p_k and p_{k+1}. Then,

$$\begin{aligned} g_\delta(p_z + \delta) &= s_i^*(p_z + \delta) - s_i^*(p_z - \delta) - 2\delta\beta_{ik} \\ &= \delta s_i^{*\prime}(p_z^+) + \delta s_i^{*\prime}(p_z^-) - 2\delta\beta_{ik} + O(\delta^2) \\ &= \delta s_i^{*\prime}(p_z^+) - \delta s_i^{*\prime}(p_z^-) + O(\delta^2), \end{aligned}$$

and similarly $g_\delta(p_z - \delta) = \delta s_i^{*\prime}(p_z^-) - \delta s_i^{*\prime}(p_z^+) + O(\delta^2)$. Hence, if the derivative of s_i^* changes at p_z the continuous function g will change sign between $p_z - \delta$ and $p_z + \delta$ for δ chosen small enough. This is enough to establish the existence of $\tilde{p}_{ik}$ as required. Note that from (9), one s_i^* changing slope at p_z will imply that each $s_j^{*\prime}$ is discontinuous at p_z.

Now consider the case where one of the supply functions, say s_i^*, has a discontinuity at p_z. From Theorem 2, other supply functions $s_i^*(p_z)$ are either zero or at their capacity limits. We choose the ε_k, $k = 1, 2, \ldots, K$ so that one of the prices, say p_h, is equal to p_z. We also take p_{h-1} and p_{h+1} sufficiently close to p_h for the following inequality to hold (the left- and right-hand sides approach $s_i^*(p_z^+)$ and $s_i^*(p_z^-)$, respectively):

$$\begin{aligned} &s_i^*(p_{h+1}) - (p_{h+1} - p_h)s_i^{*\prime}(p_{h+1}) \\ &\quad > s_i^*(p_{h-1}) + (p_h - p_{h-1})s_i^{*\prime}(p_{h-1}). \end{aligned}$$

Now we need to choose ε_h so that q_{ih} is strictly between the two sides of this inequality. This is enough to ensure that for β_{ih} chosen large enough, there will be appropriate intersection points $\tilde{p}_{ih-1}$ and $\tilde{p}_{ih}$. Usually, we choose the slopes β_{jk} to match the derivatives $s_j^{*\prime}(p_k)$ so that the first equation in (14) is satisfied. But this does not happen for β_{ih}, and in fact, $s_i^{*\prime}(p_h)$ is not defined. However, observe that the equations

$$q_{jh} - (p_h - c_j'(q_{jh}))\left(\sum_{l \neq j} \beta_{lh} - D'(p_h)\right) + \lambda_{jh} - \mu_{jh} = 0, \quad j \neq i, \tag{16}$$

are easily satisfied here. When $q_{jh} = \bar{q}_j$, we can take the multiplier λ_{jh} large and positive to achieve equality in (16), and similarly if $q_{jh} = 0$, we can take μ_{jh} large and positive.

Because the number of points where at least one of $s_i^*(\cdot)$, $i = 1, \ldots, n$ is not smooth or continuous, and is finite and independent of the choice of K, the last statement in the theorem follows. □

In general, there may be many solutions to system (14) and not all of these will match a smooth SFE. However, as we refine our approximation by increasing the number of points ε_k, we can expect a limiting solution to the system to approach a true SFE.

Hence, we investigate the limit of a series of SFE for finitely many demand scenarios, when these demand shock scenarios approximate a continuous demand shock function. Let $m = 2, 3, \ldots$ and define $\varepsilon_k^{(m)} = F^{-1}((k-1)/(m-1))$, $k = 1, \ldots, m$. Let $p_k^{(m)}, q_{ik}^{(m)}, \tilde{p}_{ik}^{(m)}, \beta_{ik}^{(m)}, \lambda_{ik}^{(m)}, \mu_{ik}^{(m)}$, $i = 1, \ldots, n$, $k = 1, \ldots, m$, be a solution of (14) and let $\{s_i^{(m)}(p)\}_{i=1}^n$ be the corresponding supply functions constructed from (12). Then, we have the following theorem (when there are no jumps or discontinuities).

THEOREM 9. *If $p_k^{(m)}$, $k = 1, \ldots, m$ are global optimal solutions of* (13) *for each firm $i = 1, \ldots, n$ with the supply functions $\{s_j^{(m)}(p)\}_{j=1}^n$, and $\beta_{ik}^{(m)} < \Lambda$, $k = 1, \ldots, m$, $i = 1, \ldots, n$, for some upper bound Λ independent of m, then*:

(1) *there is a subsequence of $\{s_i^{(m)}(\cdot)\}_{i=1}^n$ which converges to a continuous SFE in the sup-norm of continuous functions for the demand shock $F(\cdot)$*;

(2) *moreover, if $\{s_i^{(m)}(\cdot)\}_{i=1}^n$ converges pointwise, then $\{s_i^{(m)}(\cdot)\}_{i=1}^n$ itself converges to a continuous SFE in the sup-norm.*

PROOF. Because the slopes β_{ik}^m are between 0 and Λ,

$$|s_i^{(m)}(p) - s_i^{(m)}(p')| \leqslant \Lambda|p - p'|$$

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

for any p, $p' \in [0, \bar{p}]$. Thus, $\{s_i^{(m)}(\cdot)\}_{i=1}^n$ is equicontinuous on $[0, \bar{p}]$, and applying Arzelà-Ascoli's Theorem (see, for example, Swartz 1994) completes the proof of the convergence parts of the theorem.

Now we show that the limit functions form an equilibrium of $\{(13)\}_{i=1}^n$. Let $\{s_i(\cdot)\}_{i=1}^n$ be the limits of the sequence $\{s_i^{(m)}(\cdot)\}_{i=1}^n$ or one of its subsequences (which is denoted by the sequence itself to save notation). It is easy to see that $s_i(\cdot)$ is nondecreasing in p for each $i = 1, \ldots, n$ because $\beta_{ik}^{(m)} \geqslant 0$. We choose an arbitrary demand shock ε (with $\varepsilon = F^{-1}(r)$ for some $r \in (0, 1)$). Then, there is a choice of k as a function of m, which we denote by k rather than $k(m)$, such that $\varepsilon_k^{(m)} \to \varepsilon$ as $m \to \infty$.

Let $\pi_i(p, s_{-i}^{(m)}, \varepsilon_k^{(m)})$ denote the profit of firm i given market price p when the other firms have the supply functions $s_{-i}^{(m)} = \{s_1^{(m)}(\cdot), \ldots, s_{i-1}^{(m)}(\cdot), s_{i+1}^{(m)}(\cdot), \ldots, s_n^{(m)}(\cdot)\}$ and the demand shock is $\varepsilon_k^{(m)}$. Because $p_k^{(m)}$ is a global maximizer of $\pi_i(p, s_{-i}^{(m)}, \varepsilon_k^{(m)})$, we have

$$\pi_i(p_k^{(m)}, s_{-i}^{(m)}, \varepsilon_k^{(m)}) \geqslant \pi_i(p, s_{-i}^{(m)}, \varepsilon_k^{(m)}) \quad \text{for any } 0 \leqslant p \leqslant \bar{p},$$

for $i = 1, \ldots, n$. Letting $m \to \infty$, we have $p_k^{(m)} \to p^*$ for some p^* (if necessary, passing to a subsequence). Thus, from the continuity of π_i, and because $s_{-i}^{(m)} \to \{s_1(\cdot), \ldots, s_{i-1}(\cdot), s_{i+1}(\cdot), \ldots, s_n(\cdot)\}$ and $\varepsilon_k^{(m)} \to \varepsilon$, we know that p^* is a global maximizer for firm i's profit given the other firms' supply functions $s_i(\cdot)$ for demand realization ε for $i = 1, \ldots, n$. □

Note that when some β_{ik} are unbounded as $k \to \infty$, which means the piecewise-linear supply curves constructed from (12) become vertical for those prices p_{ik} at which β_{ik} are unbounded, then we may still have a form of convergence. Specifically, the piecewise-linear curves may converge to an SFE with discontinuities in the Hausdorff metric (i.e., the graphs of the piecewise-linear supply functions converge to the graphs of the equilibrium supply functions); see Lemma 2.2 of Anderson and Xu (2002).

4. Implementation and Examples

In this section, we will describe how the method can be implemented and give some examples of asymmetric equilibria. Briefly, the implementation involves two procedures. The first step is to formulate a sequence of nonlinear programming (NLP) problems so that the set of feasible solutions converges to the set of solutions of (14). We use this approach because it provides a convenient way to handle the complementarity constraints between λ_{ik}, μ_{ik}, and q_{ik}. It is also helpful in dealing with the strict inequalities $p_k < \tilde{p}_{ik} < p_{k+1}$. We choose the parameters in such a way that a solution to the last iteration is a solution to (14) up to the machine precision. The second step is to check whether the $\{p_k\}$ are the global optimal solutions to (13) given the constructed supply functions $s_j(p)$, $j = 1, \ldots, n$.

First, we describe the procedure to check global optimality. Let p_k, q_{ik}, $\tilde{p}_{ik}$, β_{ik}, ε_k, λ_{ik}, μ_{ik}, $i = 1, \ldots, n$, $k = 1, \ldots, K$, be the final solution of the sequence of NLP problems. Recall that p_k is already a local optimal solution of (13) for each firm and demand shock ε_k.

For each firm i, we order the points $\tilde{p}_{jk}$, $j = 1, \ldots, n$, $j \neq i$, $k = 1, \ldots, K$. These are the points at which the aggregate supply function from the other firms is nonsmooth. Now consider the problem of calculating the optimal choice of price (13) when p is restricted to lie between two neighboring points $\tilde{p}_{jk}$. Given a demand shock ε_l, and the $s_j(p)$ constructed from p_k, q_{jk}, $\tilde{p}_{jk}$, β_{jk}, this is simply a one-dimensional quadratic (concave) profit-maximization problem, and so the optimal solution of this problem can be calculated very easily. Comparing the profits of these finitely many pieces, we can find a global maximizer p_{il}^* for firm i for the given demand shock. If $p_{il}^* = p_l$ for each i and l (in our GAMS implementation, we check whether $|p_{il}^* - p_l| < 10^{-6}$), then we have reached the desired solution. Otherwise, we need to modify the objective function of the NLP problems. The objective functions used in our numerical computation perform well in the sense that if the NLP problems are successfully solved with reasonable accuracy, then this check agrees that p_k, $k = 1, \ldots, K$ are global optimal solutions of (13) for each firm and each demand shock ε_k.

Next, we describe the formulation of the NLP minimization problems. First, given K, we choose $\tilde{\varepsilon}_k = \varepsilon_{\min} + (k-1)(\varepsilon_{\max} - \varepsilon_{\min})/(K-1)$ as a discretization of the range of demand shocks in our computation.

Let $\rho > 0$. We replace the equations $\lambda_{ik}(\bar{q}_i - q_{ik}) = 0$ and $\mu_{ik} q_{ik} = 0$ in (14) with the inequalities $\lambda_{ik}(\bar{q}_i - q_{ik}) \leqslant \rho$ and $\mu_{ik} q_{ik} \leqslant \rho$ for $i = 1, \ldots, n$, $k = 1, \ldots, K$. This replacement helps the NLP solver handle the complementarity constraints, which violate a standard NLP constraint qualification. We also introduce variables ξ_{ik}, $i = 1, \ldots, n$, $k = 1, \ldots, K-1$ with

$$\xi_{ik} = \frac{p_{k+1} - \tilde{p}_{ik}}{p_{k+1} - p_k},$$

that is, $\xi_{ik}(p_{k+1} - p_k) = p_{k+1} - \tilde{p}_{ik}$. To have $\tilde{p}_{ik}$ lying strictly in (p_k, p_{k+1}), we must have $0 < \xi_{ik} < 1$. We enforce preset lower and upper bounds for ξ_{ik}, say $\xi_{ik} \in (0.0001, 0.9999)$, if not otherwise stated in our implementations.

Our computational experience suggests that using the constraints

$$\xi_{ik}(p_{k+1} - p_k) = p_{k+1} - \tilde{p}_{ik} + \rho \tag{17}$$

enables the NLP solver to more easily find a feasible solution.

To make it easier to find solutions with appropriate values of $\tilde{p}_{ik}$ (as discussed in Theorem 8), we allow ε_k to vary. But we need to ensure that the ε_k are evenly spread through $[0, \bar{\varepsilon}]$, and so we have added the following constraints to the NLPs:

$$0.25(3\tilde{\varepsilon}_k + \tilde{\varepsilon}_{k-1}) \leqslant \varepsilon_k \leqslant 0.25(3\tilde{\varepsilon}_k + \tilde{\varepsilon}_{k+1}) \quad \text{for } 1 < k < K, \varepsilon_1 = \tilde{\varepsilon}_1, \varepsilon_K = \tilde{\varepsilon}_K. \tag{18}$$

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

Our numerical experience is that there is no difficulty associated with the nonexistence of $\tilde{p}_{ik}$ when we fix $\varepsilon_k = \tilde{\varepsilon}_k$ for all k, but we have retained the distinction between ε_k and $\tilde{\varepsilon}_k$ in our base implementation.

Thus, the basic problem we have solved is the following NLPs for $\rho \to 0^+$:

$$\begin{aligned} &\underset{p,q,\tilde{p},\beta,\varepsilon,\lambda,\mu}{\text{minimize}} \quad \sum_{i,k}(\xi_{ik}-0.5)^2+\sum_k(\varepsilon_k-\tilde{\varepsilon}_k)^2 \\ &\quad \text{such that} \quad \text{(14) with aforementioned replacement of complementarity constraints (17) and (18).} \end{aligned} \tag{19}$$

We can modify the objective function to explore the existence of various SFE as seen in the rest of this section.

Our computational experience also suggests that adding $\sum_{i,k}\beta_{ik}$ to the objective function makes it easier for the NLP solver to find a feasible solution and stabilizes the solution procedure. Clearly, adding this term to the objective will make it more likely that we end up with an equilibrium with low values for $s_i(p)$, which corresponds to solutions that hold back supply.

In our implementation, we start with $\rho = 10$ and scale it down by a factor of 0.1 at each iteration; solving problem (19) a total of $N = 14$ times. For some harder problems, it may be helpful to increase the scale factor to, e.g., 0.5 (and at the same time increase the number of iterations, to say $N = 37$, to maintain a final ρ value sufficiently close to zero).

Because (19) is a large-scale problem for multiple firms and a large number of demand shocks, we need to take care with the choice of starting points for the solution of (19). In our examples, we have used the following set of starting values: $p_k = (k-1)\bar{p}/K$, $\tilde{p}_{ik} = (p_k + p_{k+1})/2$, $\varepsilon_k = \tilde{\varepsilon}_k$, $\xi_{ik} = 0.5$, $q_{ik} = \max\{0, (D(p_k)+\varepsilon_k)/n\}$.

We use the NLP solver CONOPT (Version 3.0) (Drud 1994) in the GAMS environment for all the computation in this section. Our numerical experience shows that CONOPT is more capable in solving our problems than other NLP solvers in GAMS. However, we note that CONOPT is sensitive to the choice of objective functions. In particular, for some examples, CONOPT may deem a problem infeasible for one objective function, but may find an optimal solution to the problem when a different objective function is used, even though the constraints are unchanged.

We have solved many numerical examples, but we will present only a few results due to limitations of space. In each of the examples, we assume quadratic cost functions for firms and a linear demand function. If not specified otherwise, we take $\bar{p} = 150$. We have intentionally set the price cap high enough so that this is not binding in our examples (even though this may be an important issue in some markets for electricity). For each of the calculations, we take $K = 300$.

Example 10. We begin with an example with two firms which will illustrate Theorem 7. The firms have cost functions $c_1(q) = q + 0.5q^2$, $c_2(q) = 10q + q^2$ with capacity limits $\bar{q}_1 = \bar{q}_2 = 80$, respectively, which are large enough so that no firm is constrained. The demand function is $D(p) = 0.5 - 0.5p$ with demand shock ε uniformly distributed over $[0, 100]$.

For this example, it is easy to calculate a linear SFE $s_1(p) = 0.4529(p-1)$ and $s_2(p) = 0.3279(p-10)$ for $p \geqslant 10$. This follows from the closed-form solution $s_i(p) = \alpha_i + \beta_i p$, $i = 1, 2$, with

$$\alpha_i = -C_i\beta_i, \quad \beta_i = (BW_i/2)(\sqrt{1+4/(D_iBW_i)}-1),$$

where $W_i = (2D_i + D_1D_2B)/(D_1 + D_2 + D_1D_2B)$. This is now a standard result (see, for example, Green 1999).

The affine SFE is confirmed by the numerical models (19) with the objective function $\sum_{i,k}(\xi_{ik}-0.5)^2 + \sum_{k\geqslant 30}(\beta_{ik+1}-\beta_{ik})^2$. For the price range $[1.0, 10]$, firm 2 does not supply, and so firm 1 is a monopoly with supply curve $s_1(p) = (p-1)/3$ for $1 \leqslant p < 10$. The supply function $s_1(\cdot)$ jumps from 3.0 just below $p = 10$ to 4.0764 just above $p = 10$. For prices less than 1.0, firm 1 will supply nothing. Thus, we obtain a piecewise-linear SFE which is plotted in Figure 3. Demand shocks in the interval $[7.5, 8.5764]$ all yield the same price of 10.

The other two nonlinear SFE, labelled as nonlinear equilibria 1 and 2 in Figure 3, are found using (19) with the objective functions

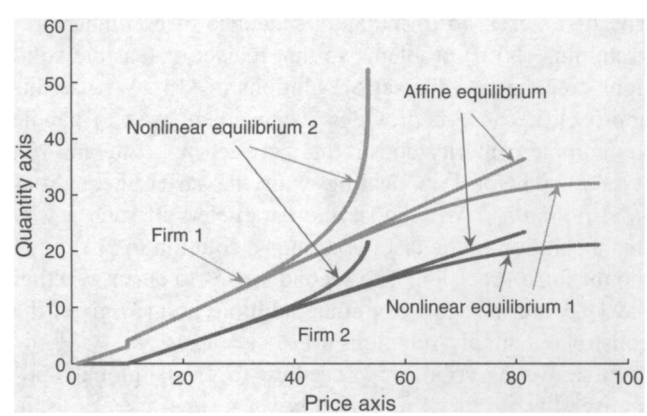
$$\sum_{i,k}(\xi_{ik}-0.5)^2+\sum_k(\varepsilon_k-\tilde{\varepsilon}_k)^2+\sum_{i,k}\beta_{ik}$$

for nonlinear equilibrium 1 in Figure 3, and

$$\sum_{i,k}(\xi_{ik}-0.5)^2+\sum_k(\varepsilon_k-\tilde{\varepsilon}_k)^2-\sum_k q_{1k}$$

for nonlinear equilibrium 2 in Figure 3, where the term $-\sum_k q_{1k}$ is used to maximize the q_{1k}s. We used the constraints $0 \leqslant \beta_{ik} \leqslant 10^6$ and $0.0001 \leqslant \xi_{ik} \leqslant 0.9999$. All three

Figure 3. Ordered multiple equilibria with a jump.



This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

equilibria have a jump at $p = 10.0$. and they can be seen to be members of an ordered family.

Note that any two equilibria, say s and $\tilde{s}$, have the property that the values for $s_i(p)$ and $\tilde{s}_i(p)$ move apart from each other as the price increases. This is a general property for problems with just two firms. To see this, observe that the equilibrium condition (2) implies

$$\frac{d}{dp}(s_1(p) - \tilde{s}_1(p)) = \frac{s_2(p)}{p - c_2'(s_2(p))} - \frac{\tilde{s}_2(p)}{p - c_2'(\tilde{s}_2(p))} > 0, \quad \text{when } p < p_{\max}$$

if $s_2(p) > \tilde{s}_2(p)$, and similarly for $(d/dp)(\tilde{s}_2(p) - s_2(p))$.

It is not hard to see that there will be a whole family of solutions in an ordered set between the nonlinear equilibria 1 and 2. This follows from Kneser's Theorem (Deuflhard and Bornemann, 2002, Theorem 2.17).

Example 11 (Intervals of Asymmetric Solutions for Symmetric Suppliers). In this example, $c_1(q) = c_2(q) = 5q + q^2$, the demand function $D(p) = 60 - 0.5p$, $\bar{q}_1 = \bar{q}_2 = 80$, $\bar{p} = 150$, and the demand shock range is $[50, 100]$. It has been shown in Klemperer and Meyer (1989) that there are only symmetric equilibria for symmetric suppliers when demand is low enough so that any supply curve will pass through $(c_i'(0), 0)$. In this example, we show the types of nonsymmetric equilibria that occur when the minimum realizable price is above the marginal supply cost at zero supply. To do so, we modify the objective function of (19) to

$$\sum_{i,k}(\xi_{ik} - 0.5)^2 + \sum_k(\varepsilon_k - \tilde{\varepsilon}_k)^2 + \sum_k(q_{2k} - q_{1k}) + (q_{21} - \tau)^2$$

with $\tau = 20$ (for equilibrium 1 in Figure 4) and

$$\sum_{i,k}(\xi_{ik} - 0.5)^2 + \sum_k(\varepsilon_k - \tilde{\varepsilon}_k)^2 + (q_{21} - \tau)^2$$

with $\tau = 25$ (for equilibrium 2 in Figure 4). The term $\sum_k(q_{2k} - q_{1k})$ has the effect of maximizing the difference between q_{1k} and q_{2k} to force an asymmetrical equilibria. We also used the bounds $0 \leqslant \beta_{ik} \leqslant 10^6$ and $0.0001 \leqslant \xi_{ik} \leqslant 0.9999$ for these solutions. We can see that these two equilibria are not in an ordered family. Theorem 7 does not apply directly here because the cost functions are the same, but even if one of the cost functions was perturbed slightly, we would have $\varepsilon_{\min}$ too large to satisfy (10).

Note that equilibria 1 has both supply functions passing through the point $(110, 52.5)$ at which they have infinite slope (because $p - c_i'(s_i(p)) = 110 - 5 - 2 \cdot 52.5 = 0$, $i = 1, 2$). This can only be part of a feasible supply function at its end point. In fact, this point is at the intersection of the locus of points where the slope is infinite ($p = 5 + 2q$) and the points which, if both supply functions are the same, correspond to the highest demand shock $\varepsilon_{\max}$ (i.e., the points $q = 0.5D(p) + 0.5\varepsilon_{\max}$, which implies $p = 320 - 4q$).

Example 12. This example has three firms and shows the uniqueness of the SFE when the minimum demand shock is low enough and only one firm does not reach its capacity limit. The cost functions of the three firms are

$$c_1(q) = 5q + 0.8q^2, \quad c_2(q) = 8q + 1.2q^2,$$
$$c_2(q) = 12q + 2.3q^2,$$

with capacities $\bar{q}_1 = 11$, $\bar{q}_2 = 8$, and $\bar{q}_3 = 55$, respectively. The demand shock ε is uniformly distributed over $[0, 50]$ and the demand function is $D(p) = 2.5 - 0.5p$. We use (19) with $0 \leqslant \beta_{ik} \leqslant 10{,}000$, $0.0001 \leqslant \xi_{ik} \leqslant 0.9999$ and the following objective function:

$$\sum_{i,k}(\xi_{ik} - 0.5)^2 + \sum_k(\varepsilon_k - \tilde{\varepsilon}_k)^2 + \sum_{i,k}\beta_{ik}.$$

The outcomes are plotted in Figure 5.

We can see a jump in quantity at the price $p = 8$ where firm 2 starts to supply from zero, but there is no jump at $p = 12$ where the third firm starts to supply from zero. This is in accordance with the result of Theorem 2. In this example, we find only one SFE in which firms 1 and 2 reach their capacity limits.

Figure 4. Multiple asymmetic equilibria not in order with symmetric suppliers.

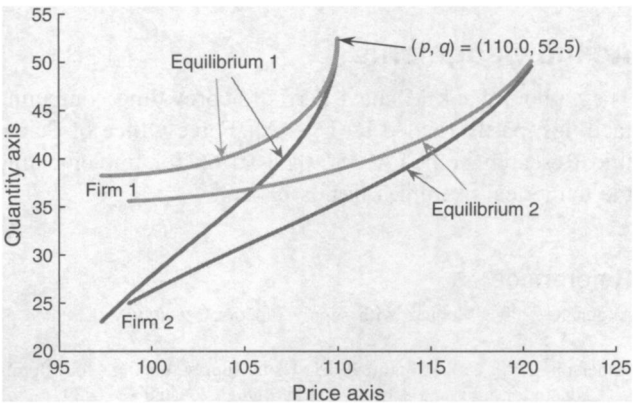
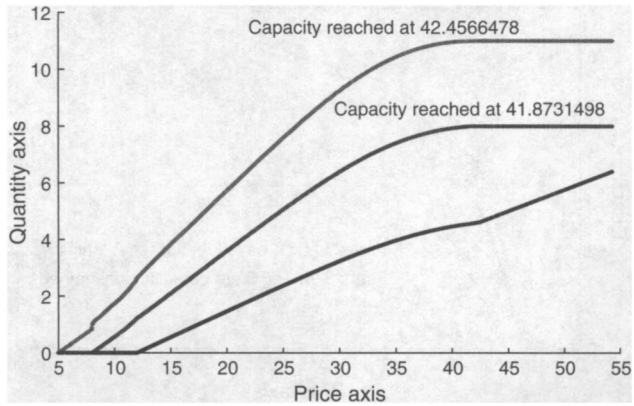


Figure 5. Locally unique SFE with capacitated suppliers and low minimal demand shock.



This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

Table 1. Cost function and capacity caps for the five generators.

	Firm 1	Firm 2	Firm 3	Firm 4	Firm 5
C_i	8.0	8.0	12.0	12.0	12.0
D_i	1.789	1.93	4.615	4.615	2.687
$\tilde{q}_i$	10.4482	9.70785	3.35325	3.3609	5.70945

Example 13. This example is for five competing firms: it is taken from Baldick et al. (2004) (see also Baldick and Hogan 2002). The cost functions and capacity limits are given in Table 1. Here we scale down the lowest demand to force all firms to have zero generation at the lowest demand shock, and then correspondingly increase the largest demand shock so that the largest demand is the same as that in Baldick et al. (2004) and Baldick and Hogan (2002).

Note that firms 3 and 4 have identical cost functions. The load duration curve is $0.8 + 34.2(1-t)$ and demand-price slope is 0.1. Hence, we have a demand function $D(p) = 0.8 - 0.1p$ and a demand shock uniformly distributed over $[0, 34.2]$. To solve this problem, we use (19) with an objective function

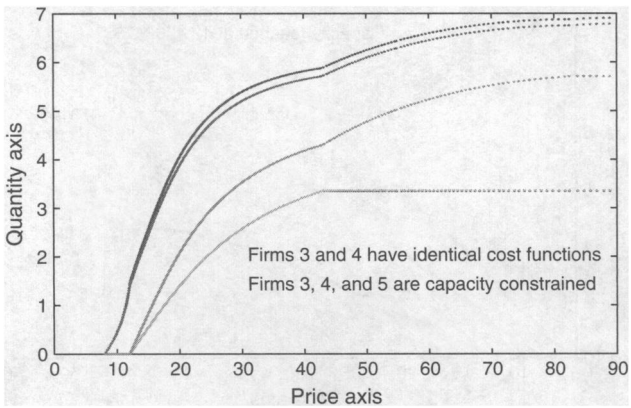
$$\sum_{i,k}(\xi_{ik} - 0.5)^2 + \sum_{k}(\varepsilon_k - \tilde{\varepsilon}_k)^2 + \sum_{i,k}\beta_{ik}.$$

We use $0 \leqslant \beta_{ik} \leqslant 100$ and $0.01 \leqslant \xi_{ik} \leqslant 0.99$ here. The capacitated SFE is given in Figure 6, where three firms are actually capacity constrained at the highest prices. This is one of a family of possible solutions.

5. Conclusions

SFE are known to be the appropriate equilibrium concept in some circumstances, but their application has been limited by difficulties in dealing with asymmetric cases. In this paper, we characterize the form of strong SFE when both capacity limits and cost functions vary from one firm to another. Our main result is that under certain conditions, SFE are ordered with at most one such equilibria in which all but one firm reaches its capacity limit.

Figure 6. Five firms with three capacity constrained.



At first sight, this result does not seem to deal with the issue of equilibrium selection. However, our numerical results present a somewhat different story. Although we cannot rule out the behaviour illustrated in Figure 1, it is very hard to find any examples for which it occurs, i.e., there seems to be almost no problems that have a family of equilibria having two or more firms not reaching their capacity constraints, at the same time as having another equilibrium in which all but one firm reaches its capacity constraints. The consequence is that for most problems for which there is an equilibrium solution, this equilibrium can be made unique simply by ensuring that the demand shocks have a large enough range (low enough so that (10) holds and high enough to cause all but one firm to reach the capacity limit). On the other hand, our experiments show that there are many problems which seem to have reasonable data, but which do not have any equilibrium solutions.

We also propose a novel scheme to approximate SFE through a discretization of the demand distribution. This results in a piecewise-linear supply function which can always be found (given a fine enough discretization) if a strong SFE exists. Moreover, we provide convergence results and demonstrate that the scheme works well in practice. In the event that there is no strong SFE, the method will fail to find a feasible solution. One difficulty that will arise with any method for finding equilibria is to distinguish between circumstances when there is no equilibrium possible and when the equilibrium exists but is hard to find.

There are a number of advantages of the scheme comparing with existing computational methods of finding equilibrium supply functions. The scheme does not need the specification of an initial point (i.e., a price-quantity pair) as is the case with ODE methods. Also, the range of realizable prices in equilibrium is determined endogenously by the demand profile and firms' profit-maximization strategies. This avoids one source of problems in methods which require a price range to be specified at the outset, which may lead to wrong conclusions.

6. Electronic Companion

An electronic companion to this paper is available as part of the online version that can be found at http://or.pubs.informs.org/ecompanion.html.

Acknowledgments

The authors thank Michael Ferris for providing computing facilities, partly funded by U.S. Air Force Office of Scientific Research grant FA9550-04-1-0192, for implementing the numerical examples in this paper.

References

Akgun, U. 2004. Mergers with supply functions. *J. Indust. Econom.* **52** 535–546.

Anderson, E. J., A. B. Philpott. 2002a. Using supply functions for offering generation into an electricity market. *Oper. Res.* **50** 477–489.

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms

Anderson, E. J., A. B. Philpott. 2002b. Optimal offer construction in electricity markets. *Math. Oper. Res.* **27** 82–100.

Anderson, E. J., H. Xu. 2002. Necessary and sufficient conditions for optimal offers in electricity markets. *SIAM J. Control Optim.* **41** 1212–1228.

Anderson, E. J., H. Xu. 2005a. Supply function equilibrium in electricity spot markets with contracts and price caps. *J. Optim. Theory Appl.* **124** 257–283.

Anderson, E. J., H. Xu. 2005b. Epsilon-optimal bidding in an electricity market with discontinuous market distribution function. *SIAM J. Control Optim.* **44** 1391–1418.

Andersson, B., L. Bergman. 1995. Market structure and the price of electricity: An ex ante analysis of the deregulated Swedish electricity market. *Energy J.* **16**(2) 97–109.

Baldick, R., W. Hogan. 2002. Capacity constrained supply function equilibrium models of electricity markets: Stability, non-decreasing constraints, and function space iterations. Working paper PWP-089 (revised), University of California Energy Institute, Berkeley, CA.

Baldick, R., W. Hogan. 2004. Polynomial approximations and supply function equilibrium stability. KSG working paper RWP05-028, Center for Business and Government, Harvard University, Cambridge, MA.

Baldick, R., R. Grant, E. Kahn. 2004. Theory and applications of linear supply function equilibrium in electricity markets. *J. Regulatory Econom.* **25** 143–167.

Borenstein, S., J. Bushnell. 1999. An empirical analysis of the potential for market power in California's electricity industry. *J. Indust. Econom.* **47** 285–323.

Day, C. J., D. W. Bunn. 2001. Divestiture of generation assets in the electricity pool of England and Wales: A computational approach to analyzing market power. *J. Regulatory Econom.* **19** 123–141.

Deuflhard, P., F. Bornemann. 2002. *Scientific Computing with Ordinary Differential Equations. Texts in Applied Mathematics*, Vol. 42. Springer-Verlag, New York.

Drud, A. S. 1994. CONOPT—A large scale GRG code. *ORSA J. Comput.* **6** 207–216.

Genc, T., S. Reynolds. 2004. Supply function equilibria with pivotal electricity firms. Working paper 1001-04, Eller College of Management, University of Arizona, Tucson.

Green, R. 1996. Increasing competition in the British electricity spot market. *J. Indust. Econom.* **44** 205–216.

Green, R. 1999. The electricity contract market in England and Wales. *J. Indust. Econom.* **47** 929–952.

Green, R., D. Newbery. 1992. Competition in the British electricity spot market. *J. Political Econom.* **100** 929–953.

Holmberg, P. 2005a. Unique supply function equilibrium with capacity constraints. Working paper 2004:20, Department of Economics, Uppsala University, Uppsala, Sweden.

Holmberg, P. 2005b. Asymmetric supply function equilibrium with constant marginal costs. Working paper 2005:16, Department of Economics, Uppsala University, Uppsala, Sweden.

Holmberg, P. 2005c. Numerical calculation of an asymmetric supply function equilibrium with capacity constraints. Working paper 2005:12, Department of Economics, Uppsala University, Uppsala, Sweden.

Kahn, E. 1998. Numerical techniques for analyzing market power in electricity. *Electricity J.* **11**(6) 34–43.

Klemperer, P. D., M. A. Meyer. 1989. Supply function equilibria in oligopoly under uncertainty. *Econometrica* **57** 1243–1277.

Laussel, D. 1992. Strategic commercial policy revisited: A supply function model. *Amer. Econom. Rev.* **82** 84–99.

Rudkevich, A. 1999. Supply function equilibrium in poolco type power markets: Learning all the way. TCA technical report 1299–1702, Tabor Caramanis and Associates.

Rudkevich, A. 2003. Supply function equilibrium: Theory and applications. *Proc. 36th Hawaii Internat. Conf. Systems Sci.*, HICSS '03. IEEE Computer Society, Washington, D.C.

Rudkevich, A., M. Duckworth, R. Rosen. 1998. Modeling electricity pricing in a deregulated generation industry: The potential for oligopoly pricing in a poolco. *Energy J.* **19** 19–48.

Swartz, C. 1994. *Measure, Integration, and Function Spaces*. World Scientific, Singapore.

This content downloaded from 129.174.182.113 on Wed, 25 Mar 2026 00:13:57 UTC
All use subject to https://about.jstor.org/terms